Dissertation Chapters 4 and 5 Comment Review

Advisor comments Ben replies and local revisions

29 October 2026

This document separates the advisor's comments and Ben's replies from my local revisions. It includes all 42 comment cards visible in the current-file review panels captured for Chapters 4 and 5, their complete displayed discussion, an item-by-item disposition, and the exact revised local passages. It is a review record, not a statement that the advisor has approved the edits.

No Overleaf files, comments, or tracked-change decisions were changed. The local chapter edits described here already exist in ProofFinal. Ben's plain-text anchors and instructions from the annotated Word copy are recorded below. The original annotated document is preserved. One possible dangling anchor remains explicitly qualified. Replies are transcribed as displayed, including their original wording. "No Ben reply captured" means no reply appeared in that captured card.

Scope Chapter 4 has 8 captured cards; Chapter 5 has 34. Historical resolved comments and the complete tracked-change history are outside this record. The short comments "huh?", "which?", and similar messages are not assigned invented anchors.

Ben ruling on FVU Ben directed me to ignore the advisor's FVU comment because his reply already addresses it. My added interpretation was removed from the chapter. No optimization recalculation or metric change is authorized or planned.

Known unfinished work A current relevant SADb table capture in the main chapter and a documented AI-rating methodology account remain open. Ben deferred the larger classification review. Hybrid-point removal is complete. The earlier claim that the rating methodology was in an appendix was removed locally because no such account was found in the current chapter sources. This removal is disclosed, not presented as satisfying the advisor's request.

Chapter 4 Comments and Responses

Comment 4 1

Advisor comment 29 October, 8:27 am where is this?

Ben reply 29 October, 11:48 am uploaded now. It's the one from yesterday where you pointed out that some of the points should be pink.

Disposition Already replied by Ben

My fix or remaining work No figure was uploaded in this local revision pass. Ben reports that the missing figure was uploaded. The original comment does not name the figure, so I have not marked it verified.

Comment 4 2

Ben's saved document instruction – restructure 30:549 paragraph.

Advisor comment 29 October, 8:28 am Tell the important result, than show. Not other way around.

Ben reply No Ben reply captured.

Disposition Implemented at Ben-confirmed anchor

My fix or remaining work Restructured the 30:549 paragraph to lead with reduced prediction error before introducing the figure.

Exact local text P01 in the revised passages section. Source: 30-torque.tex, line 549.

Comment 4 3
Ben's saved document instruction – do not change

Advisor comment 29 October, 8:31 am It seems to me a simpler explanation is that your 0 is off.

Ben reply 29 October, 12:08 pm From tolerancing? I'm not sure what you mean.

Disposition No further interpretation; Ben directive

My fix or remaining work Removed my unrequested experimental-zero hypothesis. Retained the original offset observation and Ben's unresolved reply. No fit change.

Exact local text P01 in the revised passages section. Source: 30-torque.tex, line 549.

Comment 4 4
Ben's saved document instruction – reference {eq:deltaL}

Advisor comment 29 October, 8:34 am in what equation?

Ben reply No Ben reply captured.

Disposition Implemented at Ben-confirmed anchor

My fix or remaining work Explicitly references Equation eq:deltaL.

Exact local text P01 in the revised passages section. Source: 30-torque.tex, line 549.

Comment 4 5
Ben's saved document instruction – no change

Advisor comment 29 October, 8:45 am Does FVU really make sense in this context? Having FVU values over 1 raise a flag for me. More variance unexplained than there is variance? I think it is because you aren't fitting directly to the data, you are testing models. I dunno, I'm not a stats person...

Ben reply 29 October, 12:06 pm It's a quarter of it's previous value, so that's good. I hear you, but you don't want me to go back and rerun all my optimizations and then rebuild my route predictions based on those numbers. There's something about using the tendons that makes the prediction (including pre-optimized) worse. It might be as simple as being a few mm off in measuring it as installed. I think, from experience working on bikes, that the calculation predicts the cable to be stiffer than it is because in biking we have to adjust cable tension as it fatigues.

Disposition Addressed by Ben reply and no further action

My fix or remaining work Ben explicitly instructed that his reply addresses this comment. My added FVU interpretation has been removed. The metric, objective, fitted values, and optimization calculations are unchanged. No rerun is planned.

Comment 4 6
Ben's saved document instruction – he already deleted. You didn't capture the tracked change.

Advisor comment 29 October, 8:46 am Unnecessary detail.

Ben reply No Ben reply captured.

Disposition Existing advisor deletions verified and reconciled locally

My fix or remaining work Read the 08:47 and 08:49 tracked deletions in Overleaf. Removed those same two passages from the local source. No live change was accepted or rejected.

Comment 4 7

Ben's saved document instruction – replot from appropriate matlab file reassemble with panel (A). Make sure legend does not cover data.

Advisor comment 29 October, 8:49 am Remove the hybrid points from this graph

Ben reply No Ben reply captured.

Disposition Implemented and visually checked

My fix or remaining work Removed the Approximate hybrid series from the saved MATLAB figure. Preserved apparatus panel A and every X/Y value in the remaining series. Rebuilt the legend below both panels so it covers no data. Updated caption and comparison sentence accordingly.

Exact local text P02 in the revised passages section. Source: 30-torque.tex, line 592.

Comment 4 8

Ben's saved document instruction – restructure 30:606 like you did in P01.

Advisor comment 29 October, 8:50 am Tell the important result. Then show.

Ben reply No Ben reply captured.

Disposition Implemented at Ben-confirmed anchor

My fix or remaining work Restructured the biomimetic-flexor paragraph to lead with reduced error for both tested configurations. Retained all reported metrics and the later single-BPA torque shortfall.

Exact local text P03 in the revised passages section. Source: 30-torque.tex, line 606.

Chapter 5 Comments and Responses

Comment 5 1

Ben's saved document instruction -added disposition
Disposition Line 7 text "Existing reviews organize parts of this evidence \citep{van_de_crommert_review_1998, windhorst_muscle_2007, windhorst_spinal_2022}; the application links individual findings to their sources and candidate controller roles, while distinguishing observations from modeling assumptions"

Advisor comment 29 October, 9:18 am This feels incomplete.

Ben reply No Ben reply captured.

Disposition Implemented at Ben-confirmed anchor

My fix or remaining work Replaced the incomplete review-to-app transition with the concrete tasks SADb supports: comparing preparations, locating measurements, and following sources.

Exact local text P04 in the revised passages section. Source: 50-database.tex, line 6.

Comment 5 2
Ben's saved document instruction – gave you anchor
Disposition 50:9 "following" in following groups of papers.

Advisor comment 29 October, 9:19 am better word?

Ben reply No Ben reply captured.

Disposition Implemented at Ben-confirmed anchor

My fix or remaining work Replaced following groups with exploring citation links among papers in the four-tab overview.

Exact local text P05 in the revised passages section. Source: 50-database.tex, line 8.

Comment 5 3
Ben's saved document instruction – gave you anchor
Disposition line 9 "and the table, pivot, and citation views can be used offline."

Advisor comment 29 October, 9:20 am huh?

Ben reply No Ben reply captured.

Disposition Implemented and checked against app README

My fix or remaining work Explained that the records, notes, and citation graph are embedded, allowing the Table, Pivot, and Bubble map tabs to work without internet.

Exact local text P05 in the revised passages section. Source: 50-database.tex, line 8.

Comment 5 4
Ben's saved document instruction – gave you anchor
Disposition line 9 "A separate online-search tab identifies candidate additions."

Advisor comment 29 October, 9:20 am huh?

Ben reply No Ben reply captured.

Disposition Implemented and checked against app README

My fix or remaining work Named Search (online), OpenAlex, the internet requirement, and saving candidate records for later curation.

Exact local text P05 in the revised passages section. Source: 50-database.tex, line 8.

Comment 5 5
Advisor comment 29 October, 9:25 am This is awkward. The primary deliverable is the explorer. I think this tidbit need to be stated here.

Ben reply No Ben reply captured.

Disposition Implemented; rating-method account remains separate

My fix or remaining work The opening identifies the application as primary and explains all four tabs. A documented methodology for the AI tool ratings has not been found and is not invented here.

Exact local text P05 in the revised passages section. Source: 50-database.tex, line 8.

Comment 5 6

Advisor comment 29 October, 9:32 am What exactly do you mean? Duplicates were combined into 1 record?

Ben reply 29 October, 12:57 pm Yeah, zotero duplicates.

Disposition Local wording revised

My fix or remaining work Explained that duplicate Zotero entries describing one publication were combined.

Exact local text P06 in the revised passages section. Source: 50-database.tex, line 14.

Comment 5 7

Advisor comment 29 October, 9:32 am what is a 'citation search'?

Ben reply 29 October, 12:59 pm like web of science or google scholar where you can search articles that cite an article, or articles the paper cites.

Disposition Local wording revised

My fix or remaining work Replaced the unexplained search term with backward reference tracing and forward citing-paper searches.

Exact local text P06 in the revised passages section. Source: 50-database.tex, line 14.

Comment 5 8

Advisor comment 29 October, 9:35 am huh?

Ben reply 29 October, 1:00 pm Review papers helped guide a corpus search is probably a better way to describe.

Disposition Local wording revised

My fix or remaining work Stated that reviews guided the search for primary studies.

Exact local text P06 in the revised passages section. Source: 50-database.tex, line 14.

Comment 5 9

Advisor comment 29 October, 9:36 am This is a little confusing. Why where they added and then removed?

Ben reply 29 October, 1:04 pm Because it's easier to programmatically collect all the papers in our different key zotero folders, including AARL and mine, and papers that match keywords, and then to trim them away.

Disposition Implemented using Ben’s replies

My fix or remaining work Names AARL and personal Zotero collections and keyword searches; explains initial bulk imports followed by narrowing the working base.

Exact local text P07 in the revised passages section. Source: 50-database.tex, line 16.

Comment 5 10

Ben's saved document instruction – gave you anchor
Disposition line 17, "Duplicate records and"

Advisor comment 29 October, 9:36 am Didn't we say this above?

Ben reply No Ben reply captured.

Disposition Verified redundant wording absent

My fix or remaining work Duplicate merging is described once in the collection paragraph; the selection paragraph does not repeat it.

Exact local text P06 in the revised passages section. Source: 50-database.tex, line 14.

Comment 5 11

Advisor comment 29 October, 9:38 am Classification into what?

Ben reply No Ben reply captured.

Disposition Implemented

My fix or remaining work Specifies experimental preparation, afferent class, and feedback pathway.

Exact local text P07 in the revised passages section. Source: 50-database.tex, line 16.

Comment 5 12

Ben's saved document instruction – Well his confusion makes sense. It's a tautology or whatever. Like, how else would it be classified?
Disposition line 17, "Each mechanism summary was grounded in available source text"

Advisor comment 29 October, 9:38 am I'm lost.

Ben reply No Ben reply captured.

Disposition Implemented at Ben-confirmed anchor

My fix or remaining work Removed the tautological source-grounding sentence and states what information was recorded instead.

Exact local text P07 in the revised passages section. Source: 50-database.tex, line 16.

Comment 5 13
Ben's saved document instruction – I added a reply. Please see it and update.

Advisor comment 29 October, 9:40 am I find this and odd detail and not necessary. You didn't input information when it wasn't available? God I would hope that is true. Your need to call it now makes me question your scientific integrity on other things...

Ben reply 29 October, 2:57 pm That's rude. I think this whole paragraph should be shifted from passive to active tense. Records are retained when..
outside this feedback scope are removed from the working base...
missing text is flagged for removal.
This is all true and establishes good rules. There was part of what I considered the SADb in research rabbit but then I wasn't able to find text because it would be a conference poster or something, so I flagged and removed it.

Disposition Implemented using newly read Ben reply

My fix or remaining work Read Ben's 29 October 14:57 Overleaf reply. Rewrote the paragraph in active voice, retaining scope-based removal and flagging/removing records whose source text could not be located, including ResearchRabbit conference posters.

Exact local text P07 in the revised passages section. Source: 50-database.tex, line 16.

Comment 5 14
Ben's saved document instruction – call it table tab. Clarify the four main tabs.

Advisor comment 29 October, 9:43 am What table view?

Ben reply No Ben reply captured.

Disposition Implemented

My fix or remaining work Calls it the Table tab; the opening names and explains all four main tabs.

Exact local text P05 in the revised passages section. Source: 50-database.tex, line 8.

Comment 5 15
Ben's saved document instruction – yeah, and call them tabs I think

Advisor comment 29 October, 9:43 am Perhaps you need a sentence overviewing the 'views'

Ben reply No Ben reply captured.

Disposition Implemented

My fix or remaining work Uses the app's four tab names consistently: Table, Pivot, Bubble map, and Search (online).

Exact local text P05 in the revised passages section. Source: 50-database.tex, line 8.

Comment 5 16

Ben's saved document instruction – I don't know the point of this either, or if it's still there
Disposition line 17, "Appendix~\ref{app:sadb_records} records the smaller maintenance subset separately"

Advisor comment 29 October, 9:46 am huh?

Ben reply No Ben reply captured.

Disposition Implemented at Ben-confirmed anchor

My fix or remaining work Removed the unexplained maintenance-subset aside. Appendix content and its numerical values remain unchanged.

Exact local text P08 in the revised passages section. Source: 50-database.tex, line 20.

Comment 5 17

Ben's saved document instruction – Let's put a pin in this and maybe I'll have a session comb the records later, but it's a huge job.

Advisor comment 29 October, 10:02 am Isn't this the whole point of the database? With how few papers (compared to the total number) are actually classified into afferent data and feedback, I think you should say at the top of this that you have 'started this database', as there is clearly still a lot of work to do on it.

Ben reply No Ben reply captured.

Disposition Deferred by Ben

My fix or remaining work Ben elected to defer the larger record-classification review. No corpus sweep or reclassification was performed. Existing coverage counts and limitations remain.

Exact local text P09 in the revised passages section. Source: 50-database.tex, line 63.

Comment 5 18

Ben's saved document instruction – Papers category
Disposition line 23, "Within that category"

Advisor comment 29 October, 10:02 am which?

Ben reply No Ben reply captured.

Disposition Implemented at Ben-confirmed anchor

My fix or remaining work Replaced Within that category with Within the Papers category.

Exact local text P10 in the revised passages section. Source: 50-database.tex, line 22.

Comment 5 19

Ben's saved document instruction – I don't know what this is either, but it's probably the missing callout from the categories represented on the bottom of the bubble chart layout?
Disposition line 23, "Louvain partition resolves 18 named topic clusters"

Advisor comment 29 October, 10:03 am huh?

Ben reply No Ben reply captured.

Disposition Clarified; no new callout invented

My fix or remaining work Explains that Louvain groups papers by citation connections into 18 named topic clusters for navigating the Bubble map. The existing six figure callouts remain.

Exact local text P10 in the revised passages section. Source: 50-database.tex, line 22.

Comment 5 20

Ben's saved document instruction – gave you anchor
Disposition line 23, "distributes the records, citation graph, and cluster assignments together"

Advisor comment 29 October, 10:03 am awkward

Ben reply No Ben reply captured.

Disposition Verified obsolete wording absent

My fix or remaining work The former distributes the records phrase is absent; the introduction instead explains why embedded records, notes, and graph support offline use.

Exact local text P05 in the revised passages section. Source: 50-database.tex, line 8.

Comment 5 21

Advisor comment 29 October, 10:07 am Is this a paragraph about all the views, or just the table view? Use topic sentences.

Ben reply No Ben reply captured.

Disposition Implemented

My fix or remaining work The Table-tab paragraph starts with its subject.

Exact local text P11 in the revised passages section. Source: 50-database.tex, line 24.

Comment 5 22

Advisor comment 29 October, 10:08 am You point to the map, then the next paragraph is about the table, then you go back to the map. You are jumping all over the place.

Ben reply No Ben reply captured.

Disposition Local organization revised

My fix or remaining work Placed the table description before the map description and moved the pivot description into Results. The exact revised map paragraph is included below.

Exact local text P12 in the revised passages section. Source: 50-database.tex, line 26.

Comment 5 23

Advisor comment 29 October, 10:11 am It would be helpful to have this figure in the main text.

Advisor follow up 29 October, 10:13 am Looking at the figure. I am super confused why irrelevant papers would show up in the app if this is a curated list. They seem to be 'removed from Airtable'. Again, very confused.

Ben reply No Ben reply captured.

Disposition Not fixed

My fix or remaining work The old table screenshot has not been promoted into the main chapter. Visual review found that it includes generic machine-learning papers and an older header. A relevant, current capture is still needed. The app itself was not modified.

Comment 5 24

Advisor comment 29 October, 10:17 am Is this something you toggle between or is it y and x axes? The wording here is confusing.

Ben reply No Ben reply captured.

Disposition Implemented

My fix or remaining work Names the two selectable layouts; identifies the year–citations axes and explains the spring layout used for topic landscape.

Exact local text P12 in the revised passages section. Source: 50-database.tex, line 26.

Comment 5 25

Ben's saved document instruction – gave possible anchor
Disposition not sure, it might be dangling or it might be line 29, "topic landscape based on the citation-network layout"

Advisor comment 29 October, 10:17 am huh?

Ben reply No Ben reply captured.

Disposition Possible anchor clarified; thread may be dangling

My fix or remaining work Rewrote the possible topic-landscape anchor in concrete layout terms. The document does not establish whether this is the live comment's anchor, so no thread resolution is claimed.

Exact local text P12 in the revised passages section. Source: 50-database.tex, line 26.

Comment 5 26

Ben's saved document instruction – It said something like "Its control" instead of "The user controls"

Advisor comment 29 October, 10:18 am The user?

Ben reply No Ben reply captured.

Disposition Implemented at Ben-confirmed anchor

My fix or remaining work Changed the map-control subject to The user controls.

Exact local text P12 in the revised passages section. Source: 50-database.tex, line 26.

Comment 5 27

Advisor comment 29 October, 10:18 am what do you mean by proposed?

Ben reply No Ben reply captured.

Disposition Local wording revised

My fix or remaining work Expanded the translation wording to notes on possible applications to a robot or simulation.

Exact local text P12 in the revised passages section. Source: 50-database.tex, line 26.

Comment 5 28
Ben's saved document instruction – you can check, maybe add a callout

Advisor comment 29 October, 10:20 am two more, or one more?

Ben reply No Ben reply captured.

Disposition Implemented and app depth selector checked

My fix or remaining work Verified the 1/2/3-link selector in the app source and retained callout E for the neighborhood controls. Explains that two links adds one level beyond direct connections.

Exact local text P12 in the revised passages section. Source: 50-database.tex, line 26.

Comment 5 29
Ben's saved document instruction – need to clarify when a paper cites or is cited by another record. Disposition line 29, "incoming citations"

Advisor comment 29 October, 10:21 am huh?

Ben reply No Ben reply captured.

Disposition Implemented at Ben-confirmed anchor

My fix or remaining work Replaces incoming citations with papers that cite the focus; explicitly distinguishes these from papers cited by the focus.

Exact local text P12 in the revised passages section. Source: 50-database.tex, line 26.

Comment 5 30
Advisor comment 29 October, 10:25 am I was right, you did present this database 2x. Include both references at the start of the chapter.

Ben reply No Ben reply captured.

Disposition Local wording revised

My fix or remaining work Added both presentation years and citation keys at the chapter start.

Exact local text P13 in the revised passages section. Source: 50-database.tex, line 4.

Comment 5 31
Advisor comment 29 October, 10:49 am I think before you get specific, start broad on the database, then talk about how it is helpful for designing models to be tested in simulations and robots

Ben reply No Ben reply captured.

Disposition Local discussion revised

My fix or remaining work Discussion now starts with the general value of following evidence, then explains neural-model design.

Exact local text P14 in the revised passages section. Source: 50-database.tex, line 55.

Comment 5 32

Advisor comment **29 October, 10:49 am** You can start a little more positively. Look at all the cool features!

Ben reply No Ben reply captured.

Disposition Local wording revised

My fix or remaining work Further Improvements now opens with the features already available before discussing incomplete classification.

Exact local text P09 in the revised passages section. Source: 50-database.tex, line 63.

Comment 5 33

Advisor comment **29 October, 10:51 am** This also reads like your results

Advisor follow up **29 October, 10:56 am** You need to discuss why the database is important. One method is to write some declarative sentences of use like "The database can be used to deepen a researchers knowledge of sensory feedback pathways used in locomotion and balance control". and "The database can be used to assist in designing neural models of control for locomotion." Then describe how it can be used in this way in the rest of the paragraph.

Ben reply No Ben reply captured.

Disposition Local discussion revised

My fix or remaining work Added declarative statements about learning the literature and designing neural models; moved feature descriptions into Results.

Exact local text P15 in the revised passages section. Source: 50-database.tex, line 57.

Comment 5 34

Advisor comment **29 October, 10:53 am** This paragraph reads like results.

Ben reply No Ben reply captured.

Disposition Implemented

My fix or remaining work Pivot-tab description, figure, and pathway inventory remain in Results.

Exact local text P16 in the revised passages section. Source: 50-database.tex, line 37.

Exact Revised Passages

The passages below are copied verbatim from the current local LaTeX files. Commands and citation labels are retained so that no paraphrase is mistaken for the implemented fix. Their line numbers are those of this review copy.

P01 Chapter 4

Location 30-torque.tex line 549

```
The correction terms reduce mean pinned-knee extensor torque-prediction error in both the training
and validation tests (Table~\ref{tab:ExtPinGof}). Figure~\ref{fig:ExtPin_group} compares the measured
and predicted torque across knee angle. By inspection, the zero point of the measured torque was
shifted to the right of the original model, consistent with an offset opposite in sign to the
$\chi_{0}$ term in the previous section. The BPA in Figure~\ref{fig:ExtPin_group}\textbf{C}
illustrates the pattern: the hybrid-method calculated torque correlates well with the originally
predicted torque, while the actual measured torque diverges. The fitted wrapping term $\chi_{3}$
enters the usable-length correction in Equation~\eqref{eq:deltaL}. Table~\ref{tab:ExtPinGof} reports
the improvement of the fit. Extensor-specific terms $(\chi_{0}, \chi_{3})$ were specified while
stiffness terms $(\chi_{1}, \chi_{2})$ were kept the same as the previous section (see
Table~\ref{tab:AdoptedXi}).
```

P02 Chapter 4

Location 30-torque.tex line 592

```
I tested the extensor configuration on the biomimetic knee with a $\diameter$\qty{10}{\mm} BPA at
$l_{0}=\qty{51.8}{\cm}$. The measured torque fell well below the original prediction
(Figure~\ref{fig:Ext10mm_52cm}). Applying the extensor terms identified on the pinned knee improved
the prediction: at those terms, the optimized model reached an RMSE of \qty{1.1702}{\newton\meter},
an FVU of 0.6681, and a maximum absolute residual of \qty{2.2088}{\newton\meter}, against
\qty{3.0322}{\newton\meter}, 4.4856, and \qty{5.5870}{\newton\meter} for the original rigid-body
prediction.
```

P03 Chapter 4

Location 30-torque.tex line 606

```
The shared correction terms reduce torque-prediction error for both biomimetic-knee flexor
configurations (Table~\ref{tab:BioTorqueGof}). Figure~\ref{fig:FlxGrp} compares these predictions
with the measurements; the bracket frame for the $\diameter$\qty{10}{\mm} BPA is shown in Fig.
\ref{fig:bktFrame}\textbf{C}, and for the $\diameter$\qty{20}{\mm} BPA in Fig.
\ref{fig:bktFrame}\textbf{D}. Panel \textbf{A} presents the $\diameter$\qty{10}{\mm} actuator at
$l_{0}=\qty{41.5}{\cm}$ with a \qty{21}{\mm} artificial tendon ($\diameter$\qty{1.5}{\mm} Shimano
bicycle brake cable wrapped four times), pressurized to \qty{604}{\kPa}; the optimized model reached
an RMSE of \qty{2.8002}{\newton\meter}, an FVU of 0.8835, and a maximum absolute residual of
\qty{4.1201}{\newton\meter}. Panel \textbf{B} presents the $\diameter$\qty{20}{\mm} actuator at
$l_{0}=\qty{41.5}{\cm}$ with a \qty{14}{\mm} artificial tendon made out of the same brake cable,
wrapped three times so that six cable segments share the load, each wrap contributing two tendon
lengths and cross-sectional areas. Panel \textbf{B} was tested at \qty{620}{\kPa}, for which the
optimized model reached an RMSE of \qty{2.2070}{\newton\meter}, an FVU of 0.0748, and a maximum
absolute residual of \qty{3.7320}{\newton\meter}.
```

P04 Chapter 5

Location 50-database.tex line 6

```
SADb addresses the need to recover measured afferent properties, feedback pathways, and experimental
conditions from a dispersed literature. Reviews provide overviews of spinal and proprioceptive
feedback \citep{van_de_crommert_review_1998, windhorst_muscle_2007, windhorst_spinal_2022}. SADb
brings individual studies into a searchable collection so that a researcher can compare experimental
preparations, locate measurements relevant to a candidate feedback pathway, and follow each record to
its source.
```

P05 Chapter 5

Location 50-database.tex line 8

```
The primary deliverable is SADb Explorer, a browser application containing 1,393 literature and
related records. It has four main tabs (Figure~\ref{fig:sadb_clusters}, callout~(A)): Table searches
individual records, Pivot compares classifications, Bubble map explores citation links among papers,
and Search (online) finds candidate additions through OpenAlex. The records, notes, and citation
graph are embedded in the application, so the first three tabs can be used without an internet
connection. Search (online) requires an internet connection and lets the user save candidate records
for later curation. The application preserves both the records retained for active curation and
archived records from the collected material.
```

P06 Chapter 5

Location 50-database.tex line 14

```
This was a curated research collection, rather than a registered systematic review or an exhaustive
search of every relevant publication. I first collected records from selected AARL and personal
Zotero collections used for previous papers on locomotion and postural balance, together with a few
records already curated in Airtable by my advisor, Dr. Hunt. Duplicate Zotero entries describing the
same publication were combined into a single record using DOI and title matching. I expanded the
collection through keyword searches, references cited by relevant papers, and searches for subsequent
papers citing them. Reviews of spinal and proprioceptive feedback guided the search for primary
studies \citep{van_de_crommert_review_1998, windhorst_muscle_2007, windhorst_spinal_2022}.
```

P07 Chapter 5

Location 50-database.tex line 16

```
I initially imported the broader Zotero collections as groups, then narrowed the working base to
sensory feedback in locomotion and posture. I retained records reporting sensory-afferent
measurements, investigating feedback in locomotion or posture, or modeling a relevant sensory
pathway. I removed records outside this scope from the working base. I also flagged records whose
source text I could not locate, including conference posters found through ResearchRabbit, and
removed them from that base. The application preserves the broader collection as archived records,
which users can distinguish with the archive filter. For retained papers, I recorded experimental
preparation, afferent class, feedback pathway, and the mechanism described by the study.
```

P08 Chapter 5

Location 50-database.tex line 20

```
The application contains 1,393 entries in five record categories: 943 Papers, 247 Models, 157 Review
Papers, 42 Feedback pathways, and 4 Recorders. The searchable table includes the complete 1,393-entry
collection. The Pivot and Bubble map tabs use the 943 entries in the Papers category. Archived
records remain searchable in the application.
```

P09 Chapter 5

Location 50-database.tex line 63

```
The application combines searchable notes, classification comparisons, and citation navigation,
providing a foundation for further curation. Classification remains incomplete within the
application's 943-entry Papers category: 95 carry an afferent classification, and 119 link to named
pathways. Completing these fields would improve comparisons across preparations and help expose where
a proposed pathway depends on a small or indirect body of evidence. Missing source text should remain
explicit until the paper can be checked; model and review records should link to the primary
experiments supporting their assumptions.
```

P10 Chapter 5

Location 50-database.tex line 22

```
Within the application's Papers category, containing 943 entries, 833 carry curation notes, 606 carry
animal-preparation tags, 316 carry animal-study assessments, 119 link to feedback pathways, 95 carry
an afferent classification, and 363 carry robot-or-simulation translation notes. DOI identifiers are
present for 892 papers, and 318 have attached-PDF indicators. Within the Papers category, 730 papers
are connected by 11,130 directed citation pairs. The Louvain algorithm groups these papers by their
citation connections into 18 named topic clusters, which help users navigate related studies in the
Bubble map tab.
```

P11 Chapter 5

Location 50-database.tex line 24

```
The Table tab combines text search with filters for preparation, pathway, afferent class, record
type, archive status, year, and note or PDF availability (Figure~\ref{fig:sadb_explorer} in
Appendix~\ref{app:sadb_records}). Selecting a record opens its source information, curation notes,
and notes on how the reported findings could inform a robot or simulation. The Search (online) tab
queries OpenAlex for candidate additions and marks DOIs already represented in SADb.
```

P12 Chapter 5

Location 50-database.tex line 26

```
The Bubble map tab offers two selectable layouts (Figure~\ref{fig:sadb_clusters}). The year--
citations layout places publication year on the horizontal axis and citation count on the vertical
axis. The topic landscape places papers using a spring layout of the citation network, so position
reflects citation connections rather than publication year or citation count. The user controls the
layout, display style, record grouping, citation-neighborhood depth, and click action
(Figure~\ref{fig:sadb_clusters}, callout~(B)). When the user selects a paper, the application
identifies it as the focus in the left record area (Figure~\ref{fig:sadb_clusters}, callout~(C)),
which exposes its curation note and notes on possible applications to a robot or simulation
(Figure~\ref{fig:sadb_clusters}, callout~(D)). The neighborhood selector displays papers up to one,
two, or three citation links from the focus, following either papers cited by the focus record or
papers that cite it (Figure~\ref{fig:sadb_clusters}, callout~(E)). A two-link neighborhood adds one
level beyond the directly connected papers. Directly connected papers retain color, while papers
reached through additional links are gray. For example, in the illustrated Grillner (1985)
neighborhood, 57 in-corpus papers cite the focus, the focus cites 13 in-corpus papers, and 555 papers
lie within two links. Green triangles identify papers that cite the focus record, and red circles
identify papers cited by it.
```

P13 Chapter 5

Location 50-database.tex line 4

```
The Sensory Afferent Database (SADb) is a curated literature resource for studying sensory feedback
in locomotion and posture. It connects papers to afferent classes, experimental preparations,
feedback pathways, and computational models. I developed it within the AARL's NeuroNex Communication,
Coordination, and Control in Nervous Systems (C3NS) effort. The resource and the evaluation of AI-
assisted literature curation were presented at Neuroscience 2023 and 2024, respectively
\citep{bolen_sensory_2023,bolen_using_2024}.
```

P14 Chapter 5

Location 50-database.tex line 55

```
SADb helps researchers connect an account of sensory feedback to the experiments and models
supporting it. Searchable notes, linked records, and citation neighborhoods provide complementary
routes through a literature spread across preparations and disciplines. A reader can begin with a
familiar paper, follow its references and later citing studies, and compare the conditions under
which a feedback mechanism was observed. This makes the resource useful for learning a research area
and for identifying questions that require a closer reading of the primary evidence.
```

P15 Chapter 5

Location 50-database.tex line 57

```
The database also supports the design of neural models for locomotion and balance. Its afferent and
pathway records connect candidate sensory signals, connection signs, and phase dependence to cited
experiments (Tables~\ref{tab:sadb_afferents} and~\ref{tab:sadb_pathways}). A modeler can use these
links to distinguish a measured response from an assumption needed to implement it. Model records
provide further starting points through two-level rhythm generation and pattern formation
\citep{rybak_modelling_2006}, muscle-reflex walking \citep{geyer_musclereflex_2010}, and the
laboratory walker lineage \citep{deng_neuromechanical_2019}. For the controllers in
Chapter~\ref{ch:sns}, these records inform pathway selection; receptor dynamics, gains, delays, and
species-specific assumptions still require explicit selection and testing.
```

P16 Chapter 5

Location 50-database.tex line 37

```
The Pivot tab counts papers by two selected dimensions, including topic cluster, decade, primary
author, animal preparation, feedback pathway, and afferent classification.
Figure~\ref{fig:sadb_pivot} crosses afferent classes with animal preparations: each cell counts
papers carrying both tags. Rows and columns can be ordered alphabetically or by count. Selecting a
cell opens the corresponding records in the Table tab, and a return control restores the pivot. The
pivot is computed over the 943 entries in the Papers category, including archived entries, rather
than all 1,393 records. Because a paper can carry several animal or afferent tags, it can contribute
to several cells. The counts describe the distribution of curated records, not independent
experimental observations or the strength of evidence for a pathway.
```

Reverse Pulley Revisions Requested by Ben

Image text Removed both bottom-left explanatory text blocks.

Caption Removed the last two sentences. The full remaining caption is reproduced below.

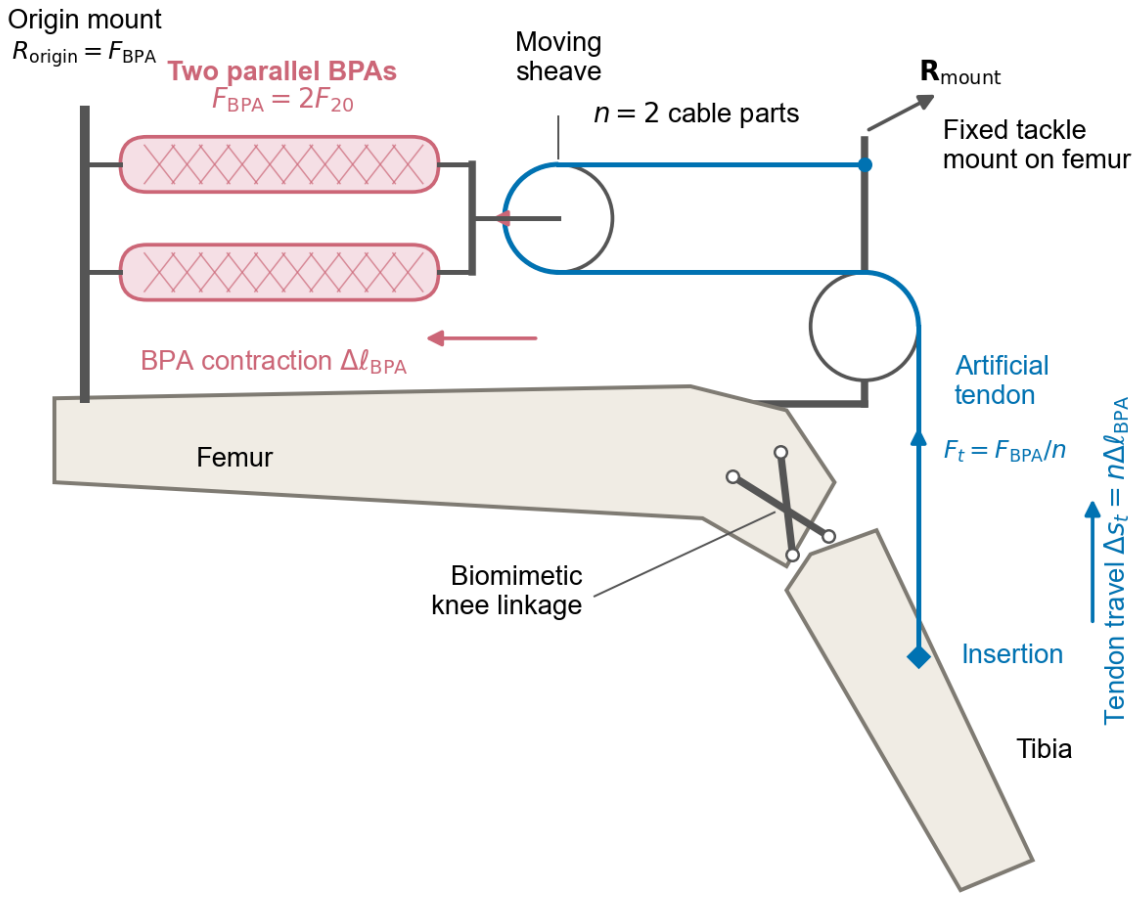
Knee mechanism Crossed the two links by connecting each proximal pin to the opposite distal pin. This remains a schematic, not a dimensioned reconstruction of the knee.

Force notation The illustrated pair is labeled F_BPA = 2 F_20. F_20 is the force of one 20 mm BPA at the stated pressure and contraction.

Reeving notation n counts cable parts supporting the moving block. It is not simply the number of pulley wheels.

Force and travel F_t = F_BPA / n and transmitted tendon travel = n times BPA contraction.

Reaction force R_mount is the support reaction at the fixed tackle mount; the BPA origin mount has a separate reaction. Direction matters, so the reaction is given as a vector and a magnitude in the chapter text.



Remaining caption Example of two parallel BPAs driving a common moving sheave. Two cable parts support the moving sheave, and a fixed sheave redirects the artificial tendon to its tibial insertion. The origin and fixed tackle mounts attach to the femur.

Force Travel and Reaction Relations

Total input force for the illustration: $F_{BPA} = 2 F_{20}$.

Transmitted tendon force: $F_t = F_{BPA} / n$.

Transmitted travel: $\Delta s_t = n \Delta \ell_{BPA}$.

Cable load at the fixed mount: $R_c = F_{BPA} u_b + (F_{BPA} / n) u_t$.

Support reaction: $R_{mount} = -R_c$.

Reaction magnitude: $|R_{mount}| = F_{BPA} \sqrt{(1 + 1/n^2 + 2 \cos \beta / n)}$.

For the drawn perpendicular cable directions and $n = 2$: $F_t = F_{20}$, $\Delta s_t = 2 \Delta \ell_{BPA}$, and $|R_{mount}| = \sqrt{5} F_{20}$.

u_b and u_t are unit vectors directed from the fixed tackle toward the moving block and toward the tendon insertion, respectively; β is the angle between them. These equations apply to the fixed tackle mount, not the total femoral support. They assume quasistatic equilibrium, frictionless pulleys, a massless cable, and rigid mounts. The cable-part count follows the actual reeving; additional redirect pulleys do not by themselves change the ratio.

Verification and Review Limits

The revised dissertation completed all five compilation passes successfully: 306 pages, no errors, no undefined references, and 1 inherited overfull-box warning(s). Fit coefficients, numerical result tables, optimization runs, published Chapter 20, and the torque-paper sources were not changed in this pass. The new force relations are ideal mechanism equations, not new measured results.

The chapter source before this session and the previous schematic are preserved in Archive/ch45_polish_20261029. The source review transcript and this ledger are preserved alongside the review files. The reverse-pulley changes and Ben-directed removal of my FVU interpretation were made during this follow-up; the document distinguishes earlier local prose revisions from remaining work.